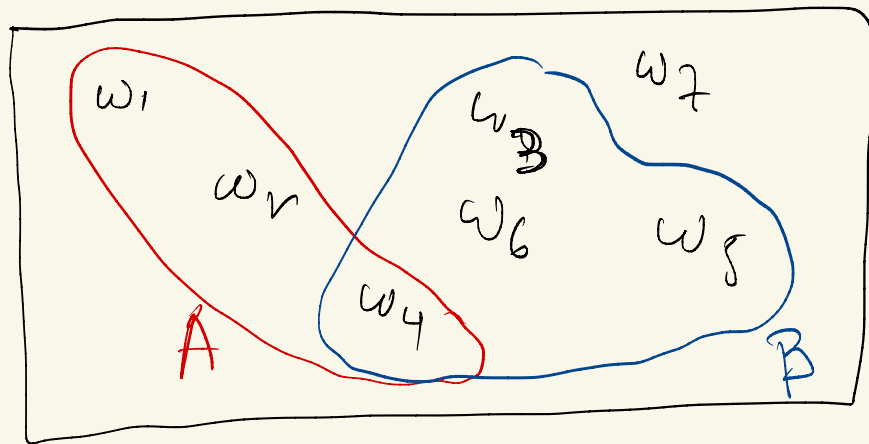


Eco 202 - Fall 2026

Class 08



Universe (Ω)

$\{ \omega_1, \omega_2, \dots, \omega_N \}$

$\emptyset = \text{empty set}$

$$A = \{ \omega_1, \omega_2, \omega_4 \}$$

$$A^c = \Omega \setminus A = \{ \omega \in \Omega : \omega \notin A \}$$

$$B = \{ \omega_3, \omega_4, \omega_6, \omega_8 \}$$

$$A \cup B = \{ \omega \in \Omega : \omega \in A \text{ or } \omega \in B \} = \{ \omega_1, \omega_2, \omega_3, \omega_4, \omega_6, \omega_8 \}$$

$$A \cap B = \{ \omega \in \Omega : \omega \in A \text{ and } \omega \in B \} = \{ \omega_4 \}$$

Def Probability (function, law, measure, map)

$$P: \{ \text{sets of } \Omega \} \longrightarrow [0, 1]$$

Sub-additivity

$$P\left[\lim_{N \rightarrow \infty} \bigcup_{j=1}^N A_j\right] = P\left[\bigcup_{j=1}^{\infty} A_j\right] = \sum_{j=1}^{\infty} P[A_j] = \lim_{N \rightarrow \infty} \sum_{j=1}^N P[A_j]$$

Recap

$$P[\Omega] = 1$$

$$P[A] \geq 0$$

Properties

$$(1) \quad \mathbb{P}[\emptyset] = 0 \quad \Rightarrow \quad \Omega \cup \emptyset = \Omega$$

$$\mathbb{P}[\Omega] + \mathbb{P}[\emptyset] = \mathbb{P}[\Omega \cup \emptyset] = \mathbb{P}[\Omega] = 1$$

$\parallel$

1

$$\Leftrightarrow \quad 1 + \mathbb{P}[\emptyset] = 1$$

$$\Rightarrow \boxed{\mathbb{P}[\emptyset] = 1 - 1 = 0}$$

$$(2) \quad \mathbb{P}[A] = 1 - \mathbb{P}[A^c]$$

$$\Omega = A \cup A^c$$

; Note $A \cap A^c = \emptyset$

thus A, A^c are disjoint

$$\Rightarrow \mathbb{P}[\Omega] = \mathbb{P}[A \cup A^c]$$

$$= \mathbb{P}[A] + \mathbb{P}(A^c)$$

$$\Rightarrow 1 = \mathbb{P}[A] + \mathbb{P}[A^c]$$

$$\Rightarrow \boxed{\mathbb{P}(A) = 1 - \mathbb{P}(A^c)}$$

③ A, B

$$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$$

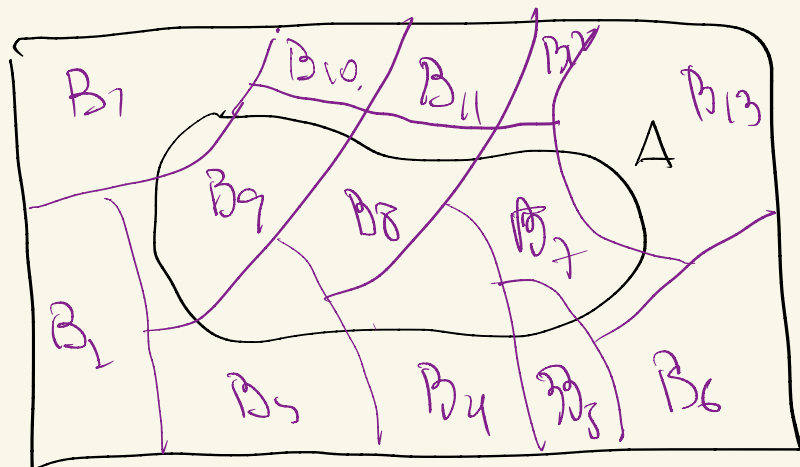
You finish this ...

④ $A \subseteq B \Rightarrow P[A] \leq P[B]$

$$B = A \cup B \setminus A$$

$$P[B] = P[A \cup B \setminus A] = P[A] + P[B \setminus A]$$

uses
additivity



$$P[A] = P\left[\bigcup_{j=1}^{13} (A \cap B_j)\right]$$

$$= \sum_{j=1}^{13} P[A \cap B_j]$$

$$A \cap B$$

$$A \cup B$$

$$A \cap B^c$$

$$A \cup B^c$$

$$(A \cap B)^c$$

$$(A \cup B)^c$$